

GeoAttend

IoT-Enabled Smart Attendance Management System

With Real-Time Geofencing, Cryptographic Device Fingerprinting, and Live Telemetry Analytics

A comprehensive project guide, operational reference, and security compliance report detailing system capabilities, architecture, and telemetry engines built from starting to final production release.

COMPREHENSIVE PROJECT REFERENCE MANUAL

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1. The Core Vision

Traditional attendance tracking systems (manual registers, RFID cards, or static QR codes) suffer from one fatal flaw: they do not guarantee physical presence. Students share QR screenshots, scan proxy RFID cards for friends, or register check-in and immediately sneak out of class (bunking).

GeoAttend is a foolproof, zero-proxy attendance and security monitoring ecosystem. It locks the student to their exact physical location, their registered mobile hardware (fingerprint signature), and their native biometric identity (Face ID/Fingerprint). It doesn't just record attendance once; it continuously monitors presence on campus during college hours, ensuring true administrative accountability.

2. System Architecture & Flow

The application enforces a rigorous multi-layered verification check for students during login and check-in:

1. Hardware Identification: Generates a stable browser fingerprint to bind the account to the phone hardware.
2. College Hours check: Verifies if operational hours (e.g. 9:00 AM - 4:00 PM) or Holiday Mode are active. Non-HOD users are rejected if closed.
3. Biometric Verification: Calls WebAuthn cryptographics to verify student biometrics.
4. Geofencing check: Validates student GPS coordinates against active class session bounds.
5. Dynamic Token check: Requires scanning a dynamic 15-second rotating QR code on the teacher's panel.
6. Campus Guard loop: Telemetry loop continues broadcasting location and battery status every 15-25 seconds.

3. A to Z Implemented Features (Modules 1 - 6)

Module 1: UI Pill Icon Badges

Replaced text stats with graphical indicators in dashboards. Battery levels use green/yellow/red pill shapes showing percentage and charging states. Pulsing green/red location dots indicate live telemetry connection status.

Module 2: Location GPS Gate

Applied geofence verification on student and teacher portals. The application blocks dashboard access with a full-screen warning if location permissions are denied, preventing location hiding.

Module 3: College Hours Auto-Lock

Configured operating hours (e.g. 9:00 AM - 4:00 PM). Student and staff dashboards lock automatically outside these hours. Includes a periodic 60-second recheck that auto-locks the page at closing time.

Module 4: Holiday Mode Override

Adds a manual holiday lockout switch on the HOD settings panel. When enabled, it overrides standard college hour gates to lock student access for the day. HOD consoles bypass this block to allow settings resets.

Module 5: Stable Browser Fingerprinting

Replaces random storage keys with stable browser hashes (combining timezone, screen color depth, language, platform, hardware concurrency). Locks accounts to physical hardware and stops proxy cache resets.

Module 6: Permanent Biometric WebAuthn Lock

Student biometrics are registered and cryptographically bound using WebAuthn. Once registered, re-registration is locked to prevent security overrides unless explicitly reset by HOD/Coordinator.

3. A to Z Implemented Features (Modules 7 - 11)

Module 7: Fake GPS & Spoofing Detection

Tracks distance velocity, accuracy anomalies (0m or >5000m), and repeated coordinates to block mock location apps. Suspicious telemetry immediately flags the student as LOCATION SUSPECT on dashboards.

Module 8: GPS Dropout Warning & Escalation

Triggers alerts if a student disables location services during active periods. Coordinators can review the warning and escalate it to HOD dashboard feeds with one click.

Module 9: Auto-Closing Sessions

Launches active sessions with a configurable timer. The server auto-closes the session once the timer runs out, preventing teachers from leaving classes open indefinitely.

Module 10: Half-Day Attendance Calculation

Accumulates checking minutes. Computes full-day ($\geq 75\%$ hours on campus), half-day (30-74% hours), or absent ($<30\%$ hours) metrics automatically.

Module 11: Student Movement Timeline

The system records telemetry logs in a new database table (movement_events) to build a continuous path history. Each heartbeat ping records state shifts: CHECKIN, IN_CAMPUS, LEFT_CLASS, RETURNED, EXITED_CAMPUS, and GPS_DROPPED. The HOD can view a complete chronological timeline for any student directly inside the dashboard.

4. Role-Based Portals

- **HOD Portal:**

View real-time campus statistics, update college geofence settings/radius/hours, toggle Holiday Mode, issue digital out-passes, reset student biometrics/devices, and view student timelines.

- **Coordinator Portal:**

Manage class rosters, execute Keypad SMS OTP check-ins, monitor GPS dropouts, and escalate security warnings to HOD.

- **Teacher Dashboard:**

Launch periods, configure auto-close timers, project dynamic 15-second rotating QR codes, and take barcode check-ins.

- **Student Console:**

Undergo GPS gates, login via WebAuthn, verify presence, scan dynamic QR, and stream active heartbeat telemetry.

5. Technical Stack

- Backend Architecture: Node.js, Express.js
- Data Storage: SQLite (highly optimized with unique indices for fast log searches)
- Security Stack: OWASP-compliant middleware, Helmet CSP headers, Express Rate Limits (DDoS guard), secure cryptographic salt/hash, IDOR protection logic
- Frontend UI: Responsive glassmorphism interface, custom CSS-drawn charts (no heavy external chart scripts), HTML5 Geolocation API, WebAuthn API

6. Why This Project Wins (For Examiners)

1. Guarantees Physical Presence (No Proxy)

Unlike standard systems where one student shares QR screenshot to check-in their friends remotely, GeoAttend requires a dynamic 15s rotating QR code combined with high-accuracy GPS telemetry and native biometric verification. Proxy attendance is mathematically impossible.

2. Solves Classroom Bunking

Entrance readers (e.g. fingerprint gate machines) only log students in at 9:00 AM. They can immediately leave campus. GeoAttend's continuous Campus Guard loop tracks their presence throughout the entire college duration, flagging exits immediately.

3. Graceful Fallbacks (Keypad/Battery-Dead)

Real-world colleges have students with keypad phones or dead batteries. GeoAttend doesn't lock them out—coordinators can trigger a secure Keypad OTP or manual registry bypass, keeping the system highly practical and realistic.

4. Complete HOD Control

Administrators are in full control. Bounds, operational hours, holiday mode, and device binding resets are all managed directly inside the dashboard without requiring backend configuration changes.